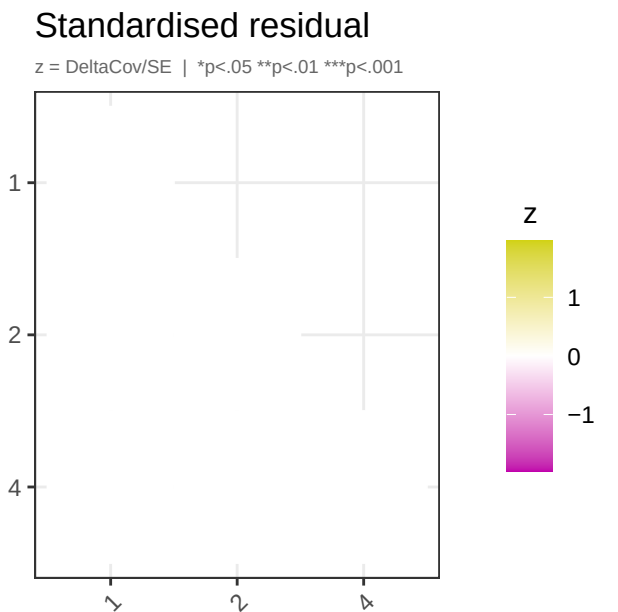
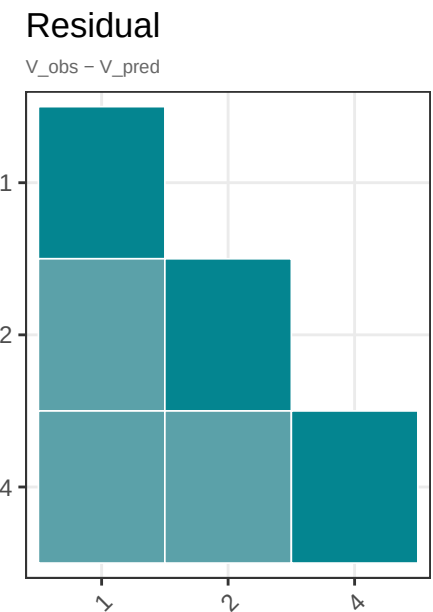
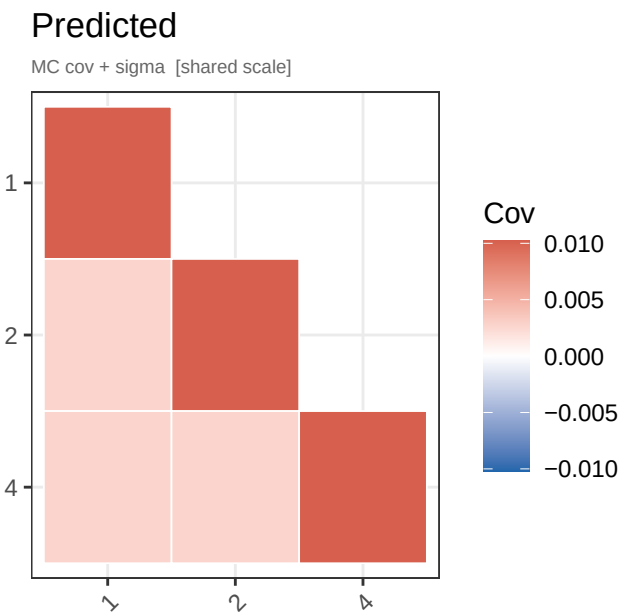
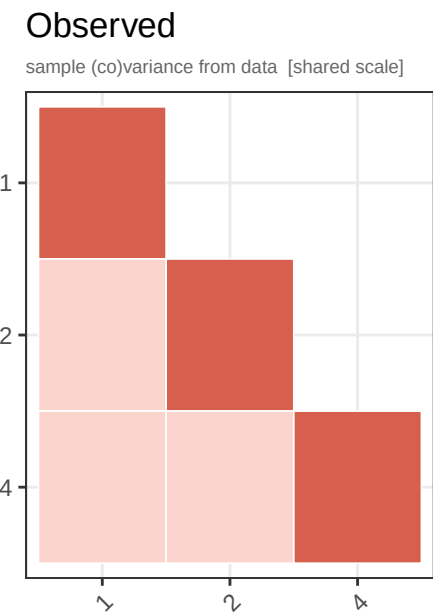


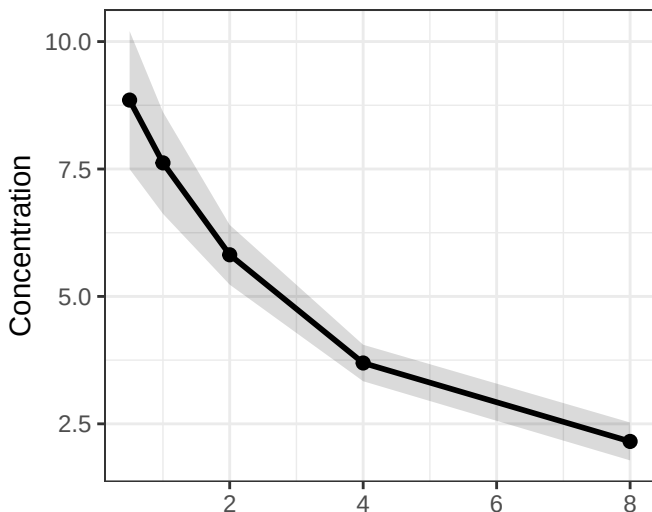
Study 's1' -- Covariance diagnostics



Study 'rat.plasma' -- Mean diagnostics

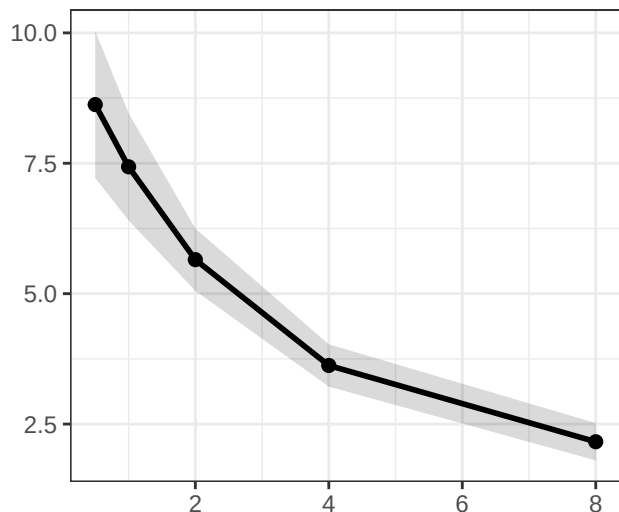
Observed

ribbon ± 1 SD [sqrtdiag(V_obs)] | shared y scale



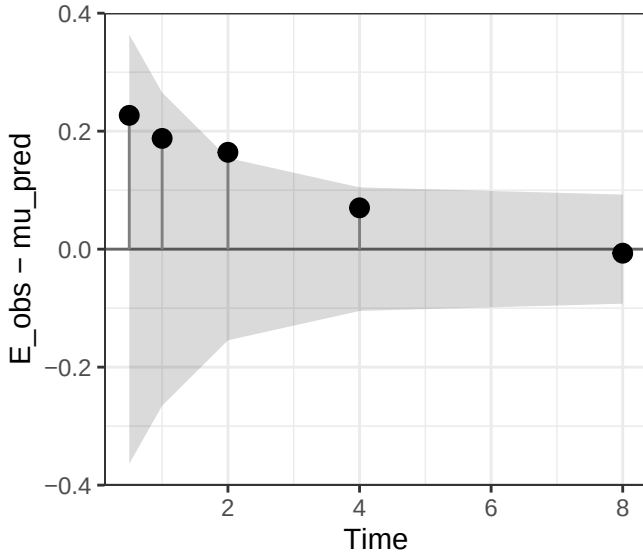
Predicted

ribbon ± 1 SD [sqrtdiag(V_pred); BSV + sigma] | shared y scale



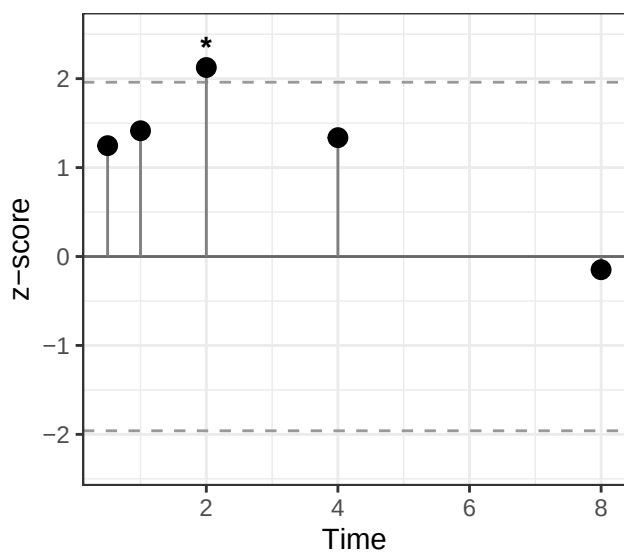
Residual

band ± 2 SE(mean) [SE = sqrt(V_pred[t,t]/n)]



Standardised residual

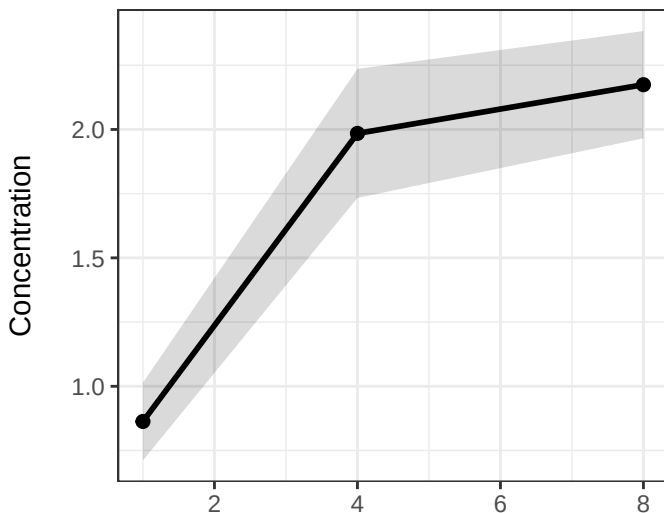
dashed ± 1.96 | $z = \text{resid}/\text{SE}$ | * $p < .05$ ** $p < .01$ *** $p < .001$



Study 'rat.tissue' -- Mean diagnostics

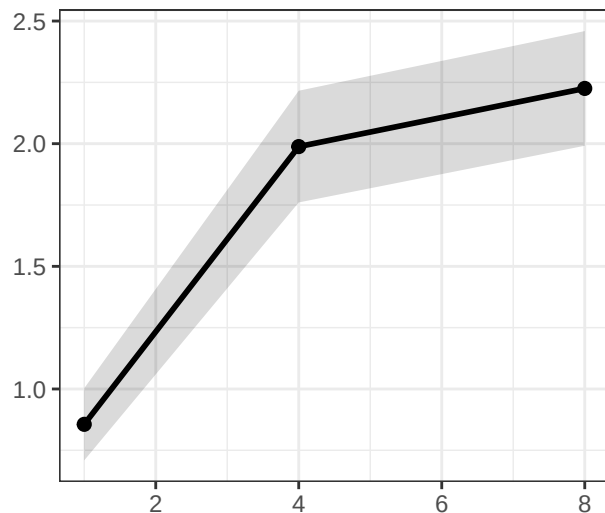
Observed

ribbon ± 1 SD [sqrtdiag(V_obs)] | shared y scale



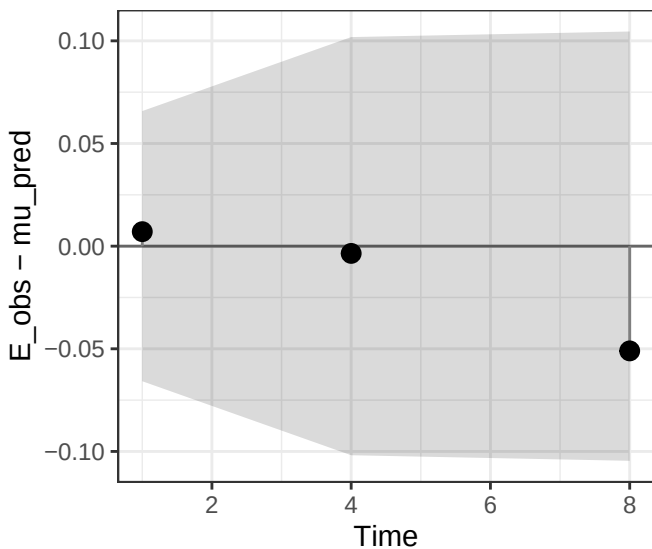
Predicted

ribbon ± 1 SD [sqrtdiag(V_pred); BSV + sigma] | shared y scale



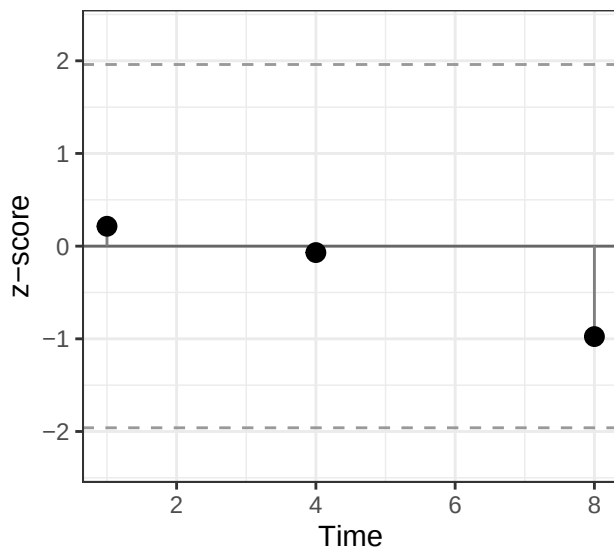
Residual

band ± 2 SE(mean) [SE = sqrt(V_pred[t,t]/n)]



Standardised residual

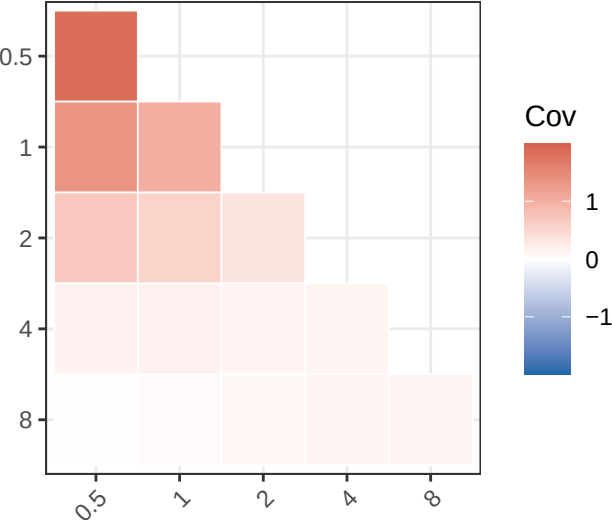
dashed ± 1.96 | z = resid/SE | *p<.05 **p<.01 ***p<.001



Study 'rat.plasma' -- Covariance diagnostics

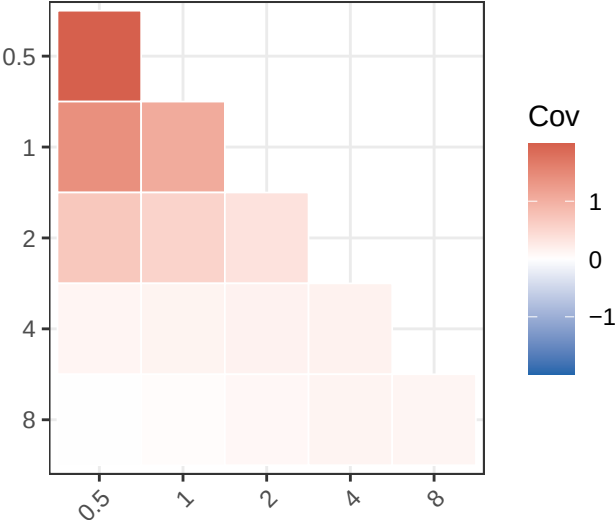
Observed

sample (co)variance from data [shared scale]



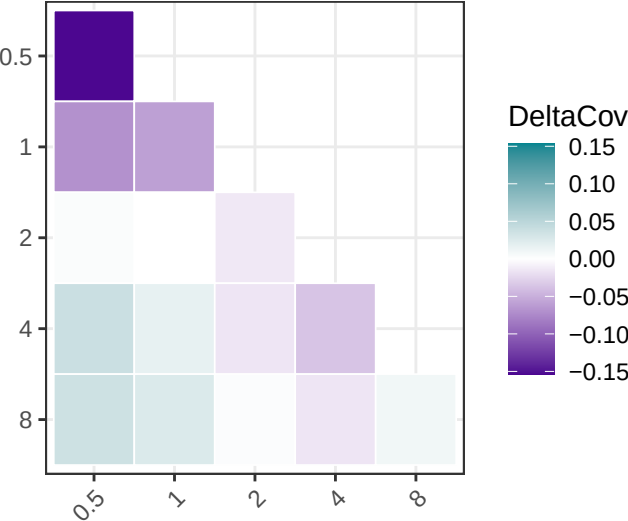
Predicted

MC cov + sigma [shared scale]



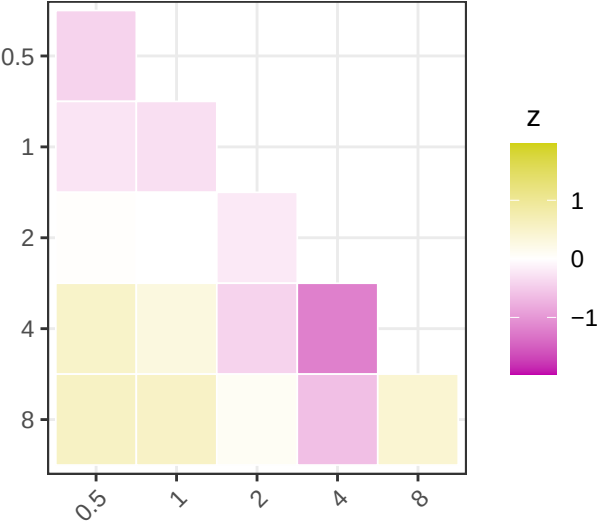
Residual

$V_{obs} - V_{pred}$

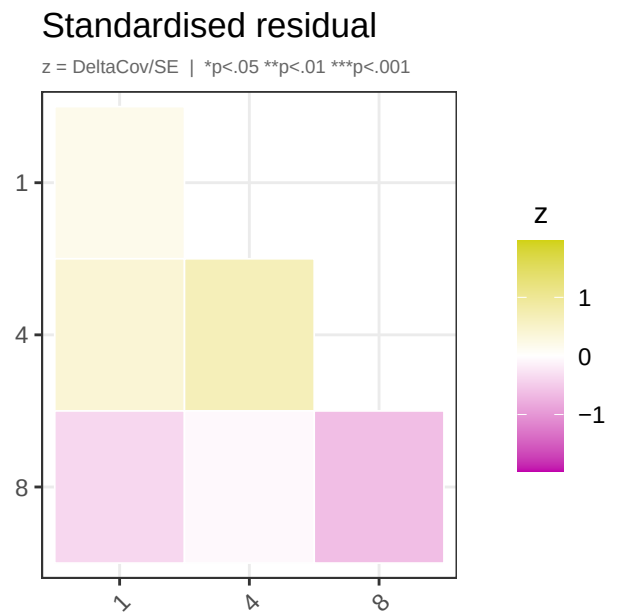
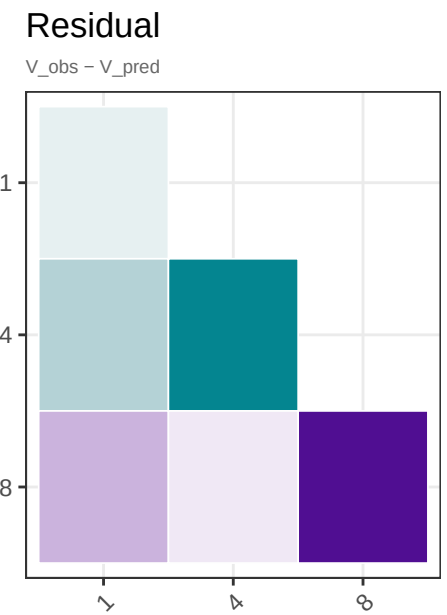
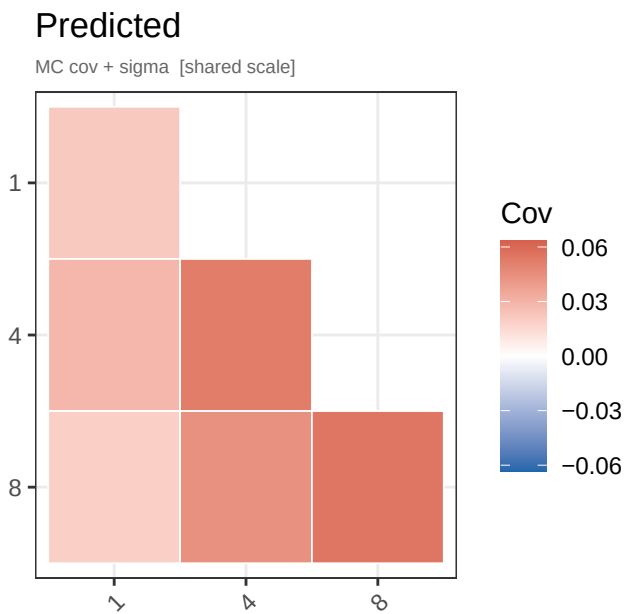
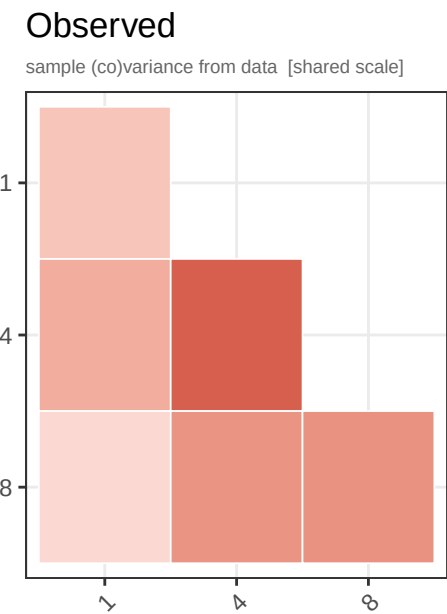


Standardised residual

$z = \text{DeltaCov} / \text{SE}$ | *p<.05 **p<.01 ***p<.001



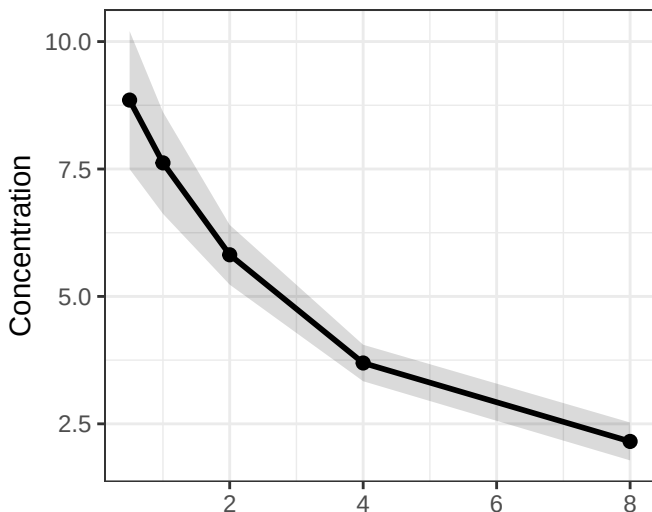
Study 'rat.tissue' -- Covariance diagnostics



Study 'rat.plasma' -- Mean diagnostics

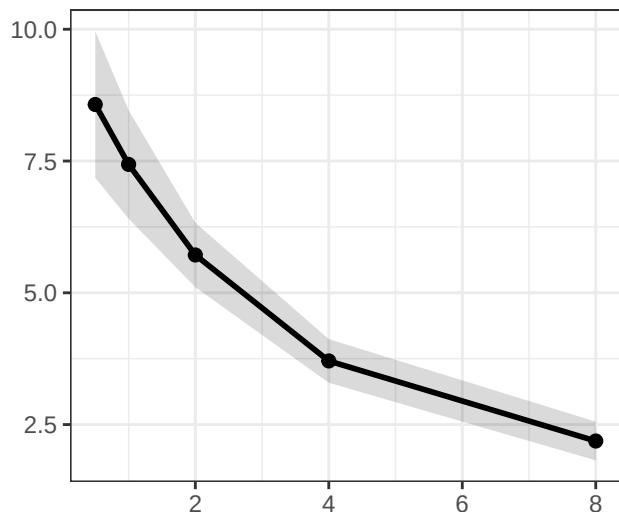
Observed

ribbon ± 1 SD [sqrtdiag(V_obs)] | shared y scale



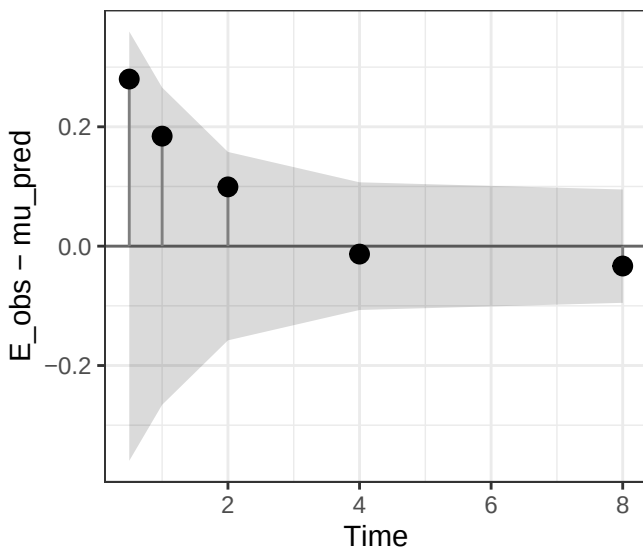
Predicted

ribbon ± 1 SD [sqrtdiag(V_pred); BSV + sigma] | shared y scale



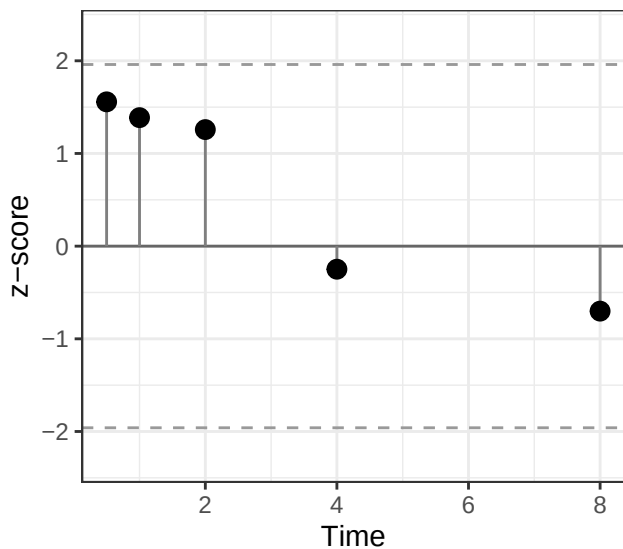
Residual

band ± 2 SE(mean) [SE = sqrt(V_pred[t,t]/n)]



Standardised residual

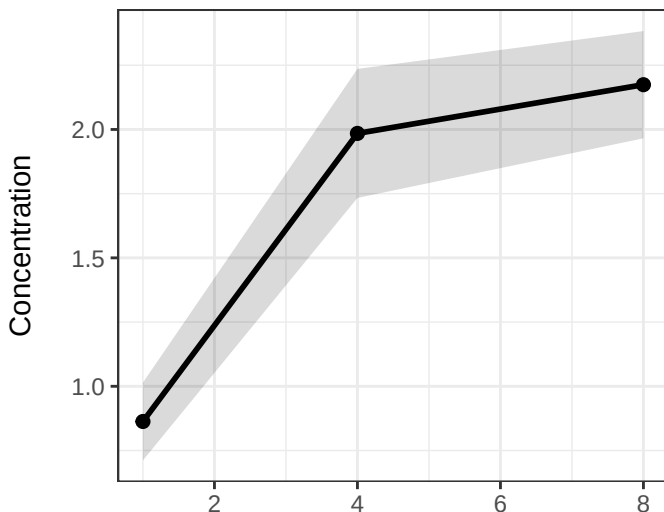
dashed ± 1.96 | z = resid/SE | *p<.05 **p<.01 ***p<.001



Study 'rat.tissue' -- Mean diagnostics

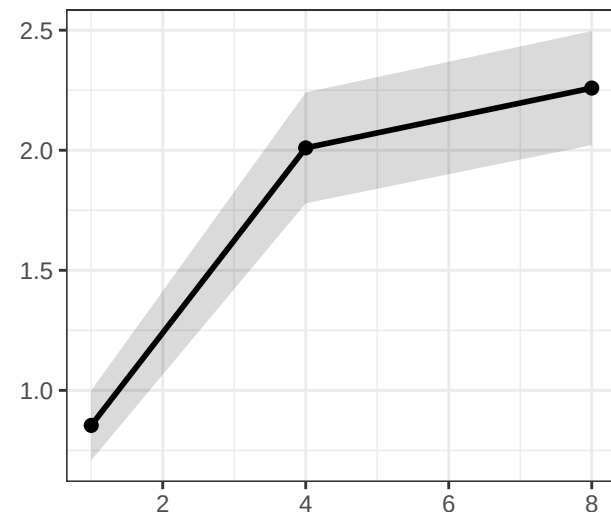
Observed

ribbon ± 1 SD [sqrtdiag(V_obs)] | shared y scale



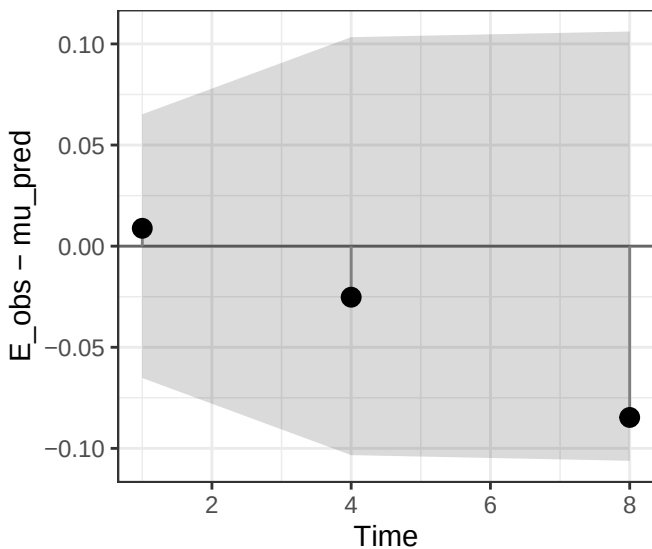
Predicted

ribbon ± 1 SD [sqrtdiag(V_pred); BSV + sigma] | shared y scale



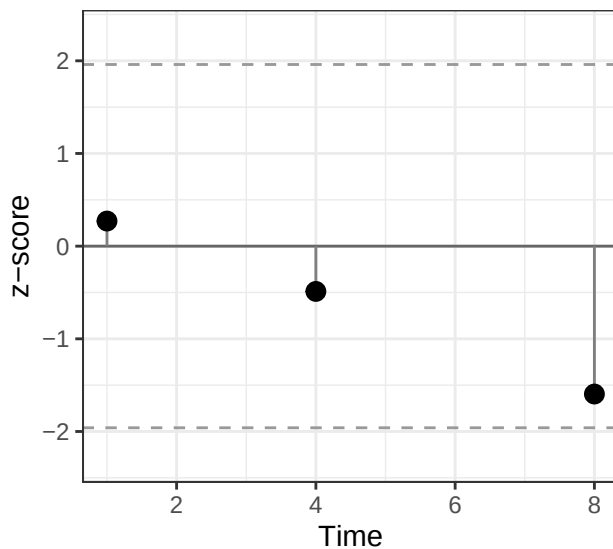
Residual

band ± 2 SE(mean) [SE = sqrt(V_pred[t,t]/n)]



Standardised residual

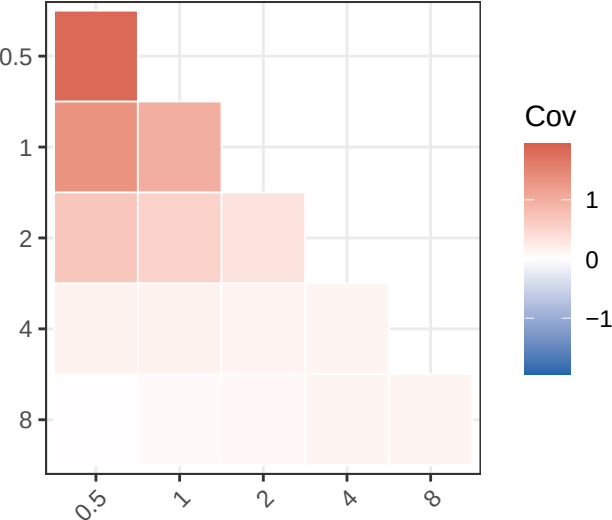
dashed ± 1.96 | z = resid/SE | *p<.05 **p<.01 ***p<.001



Study 'rat.plasma' -- Covariance diagnostics

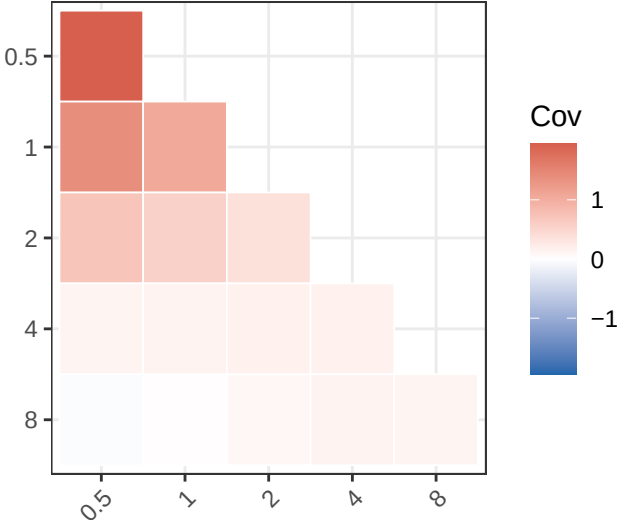
Observed

sample (co)variance from data [shared scale]



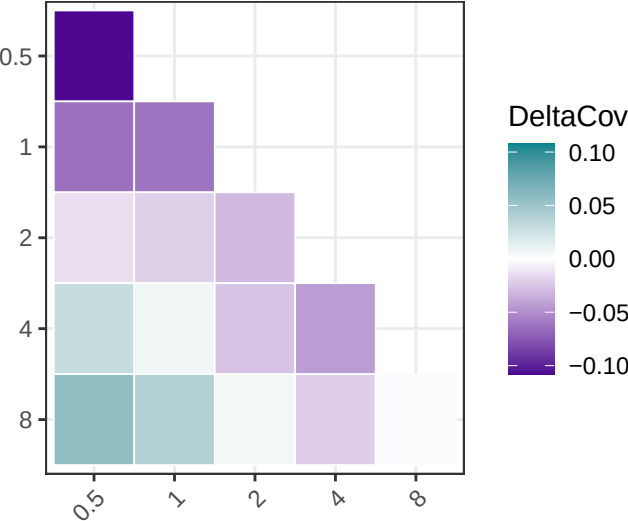
Predicted

MC cov + sigma [shared scale]



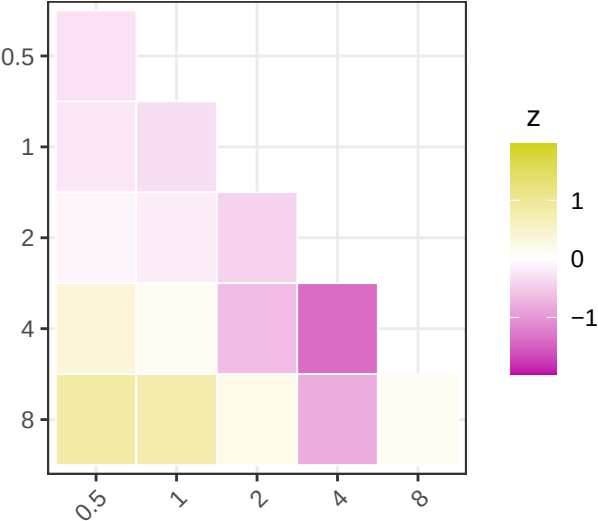
Residual

$V_{obs} - V_{pred}$

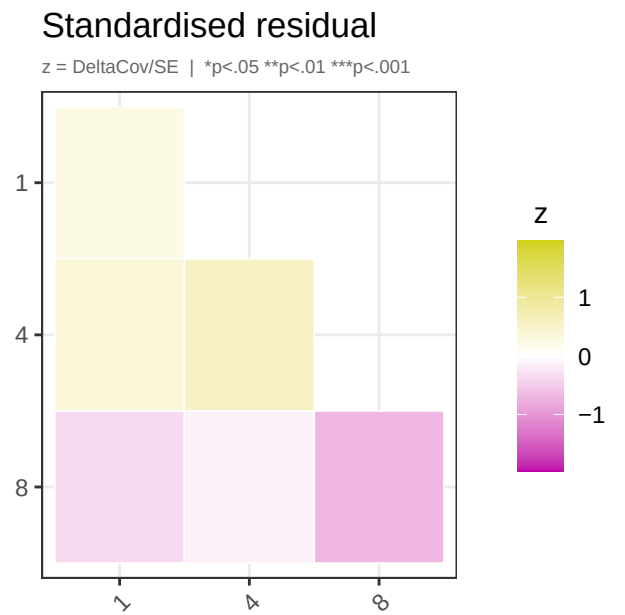
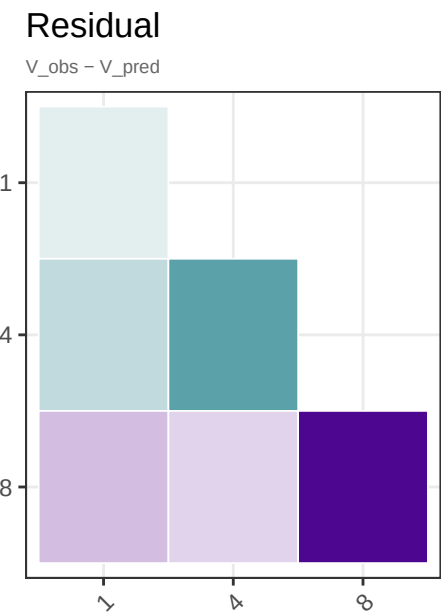
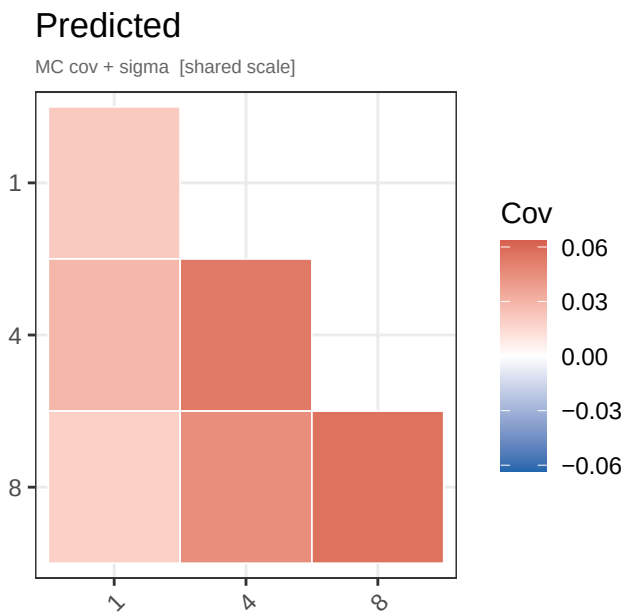
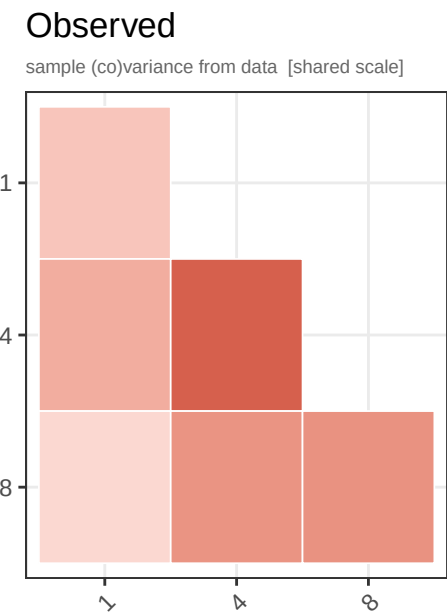


Standardised residual

$z = \text{DeltaCov} / \text{SE}$ | *p<.05 **p<.01 ***p<.001

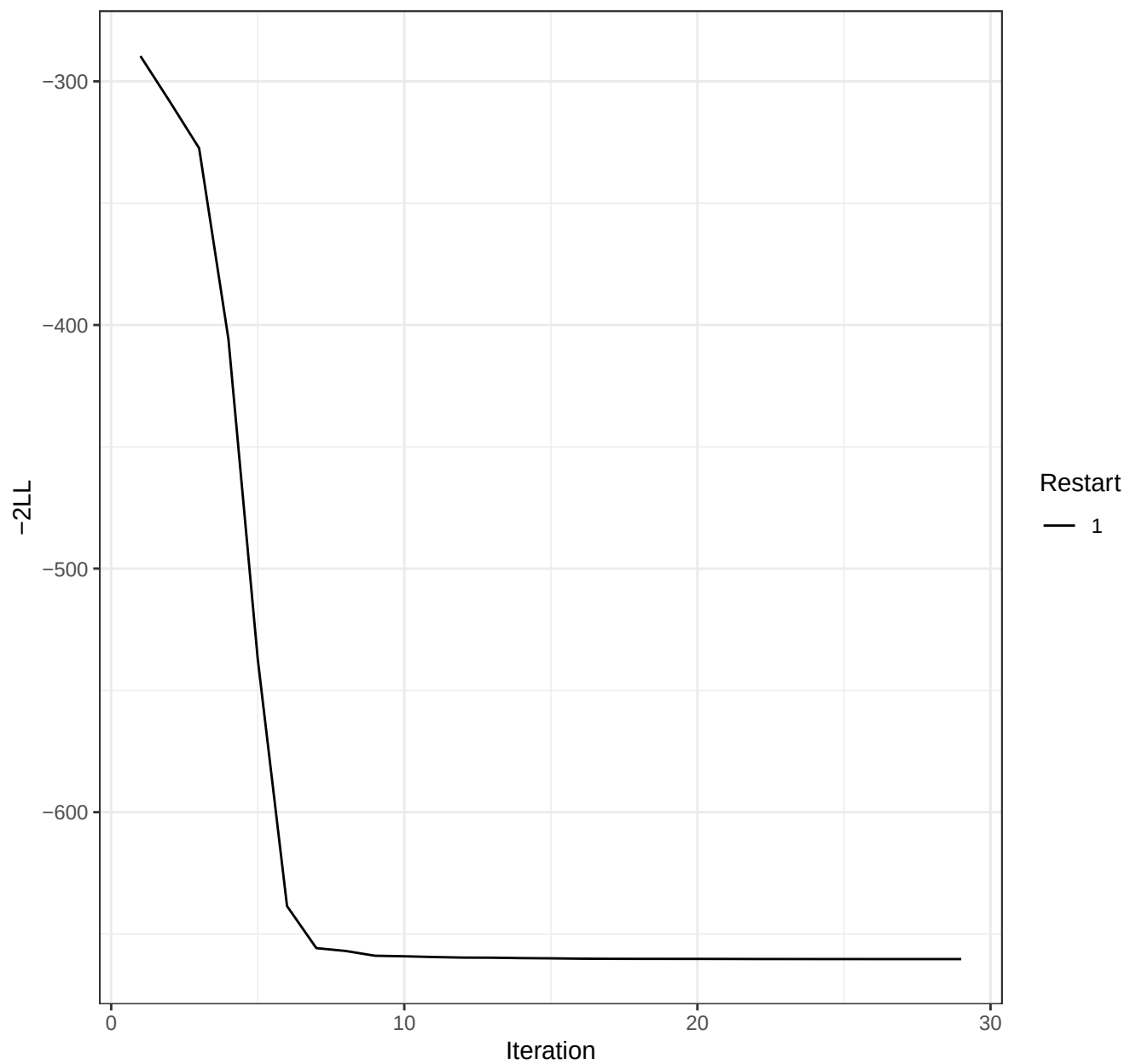


Study 'rat.tissue' -- Covariance diagnostics



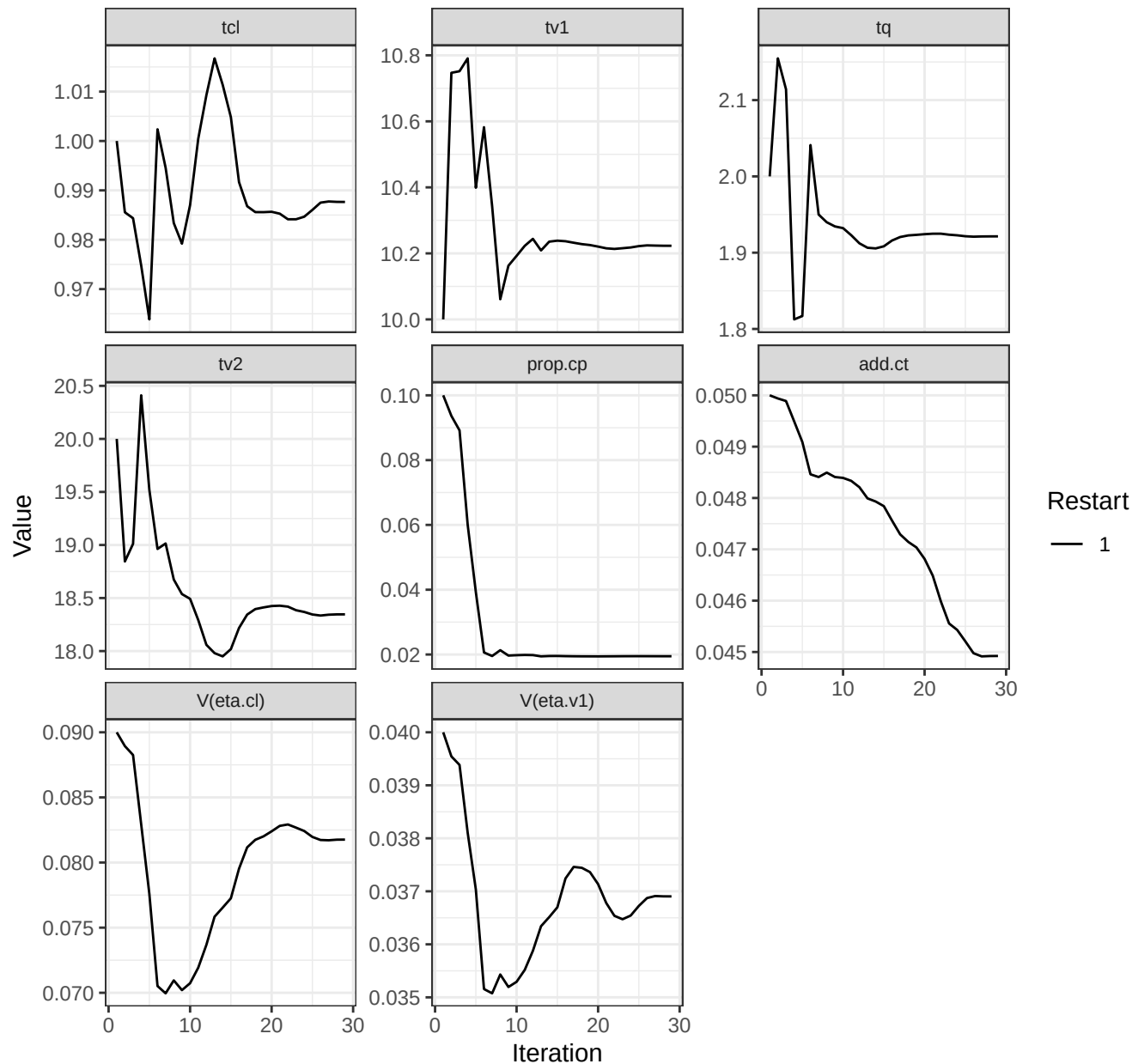
NLL trace per restart

each line = one restart; lower = better



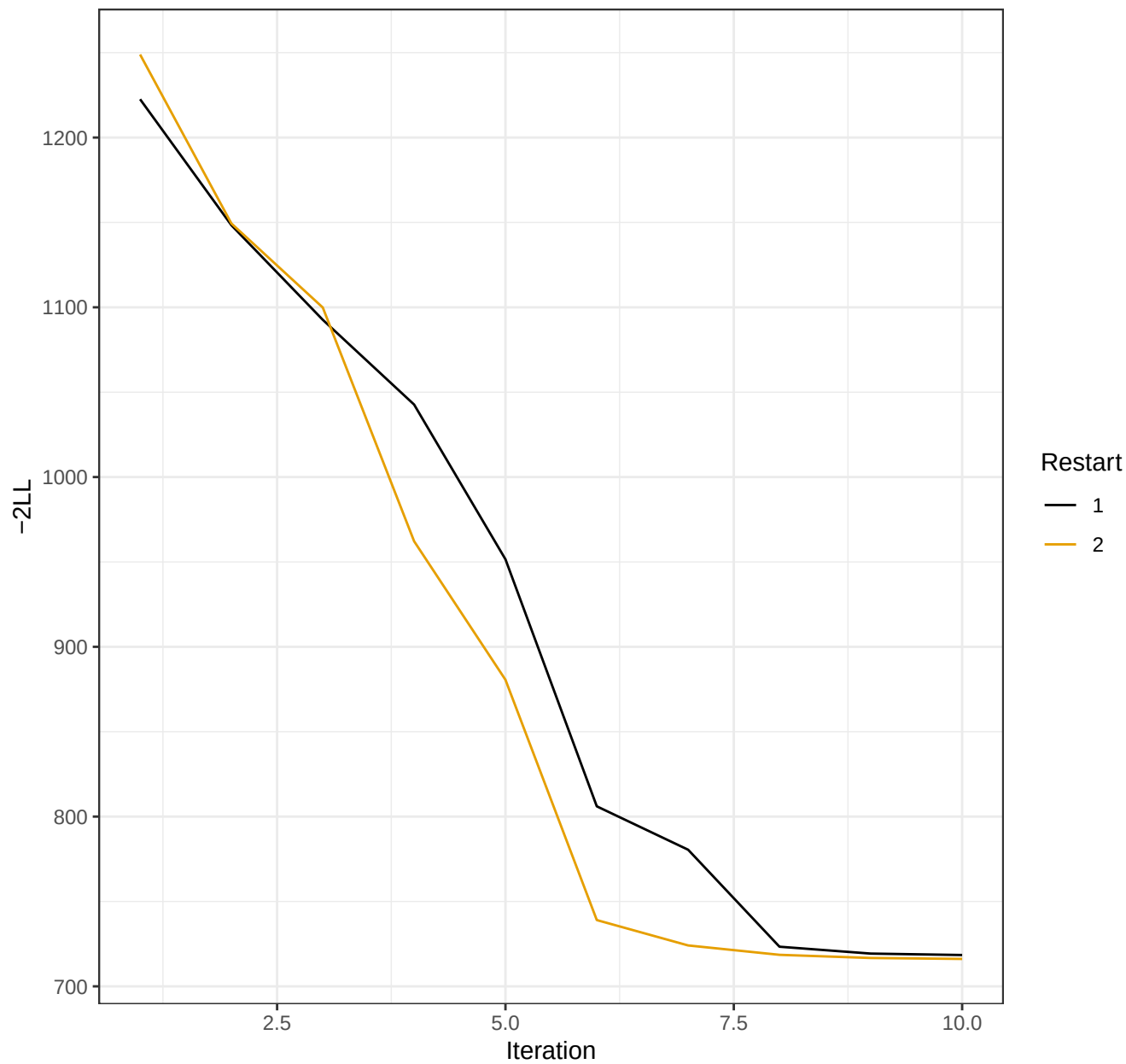
Parameter trace per restart

Natural scale; struct = back-transformed, sigma = SD, V(eta) = variance, off-diagonal = Cholesky L[i,j]



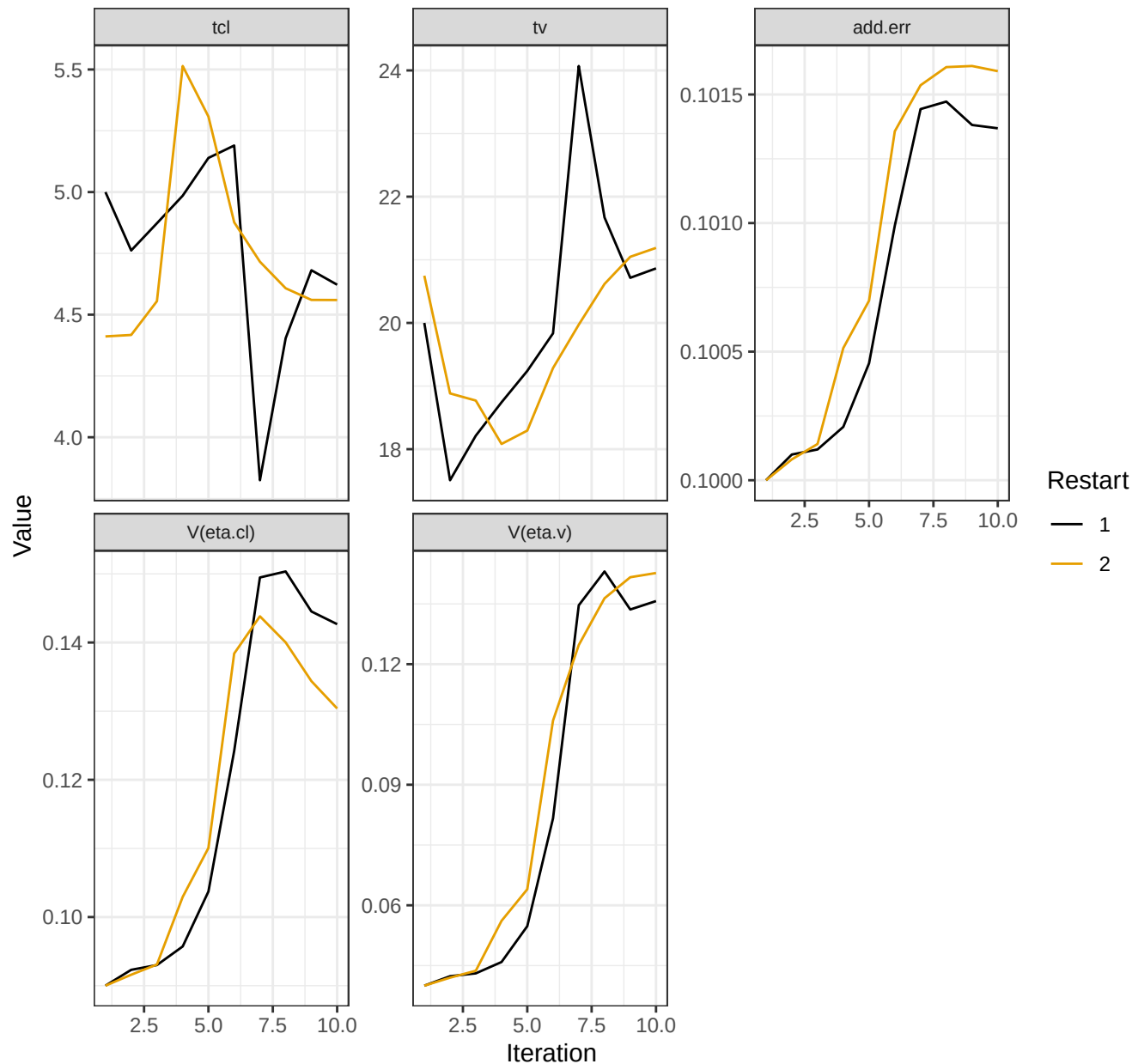
NLL trace per restart

each line = one restart; lower = better

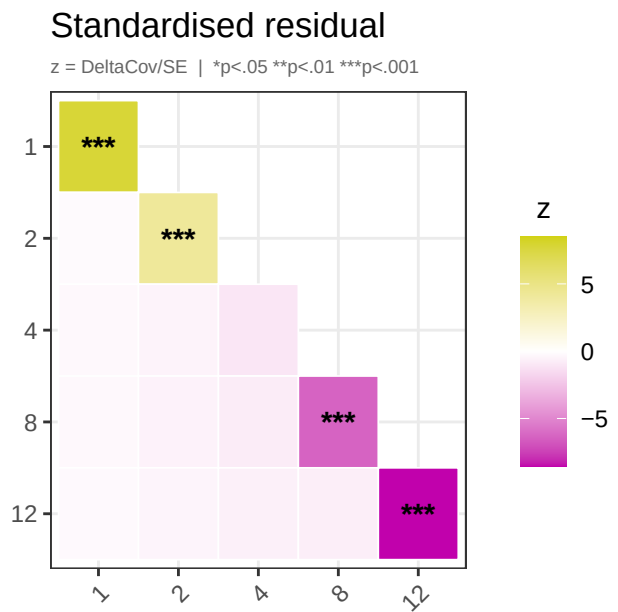
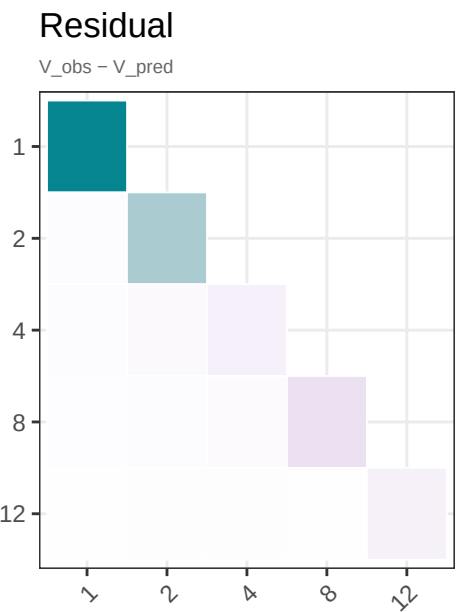
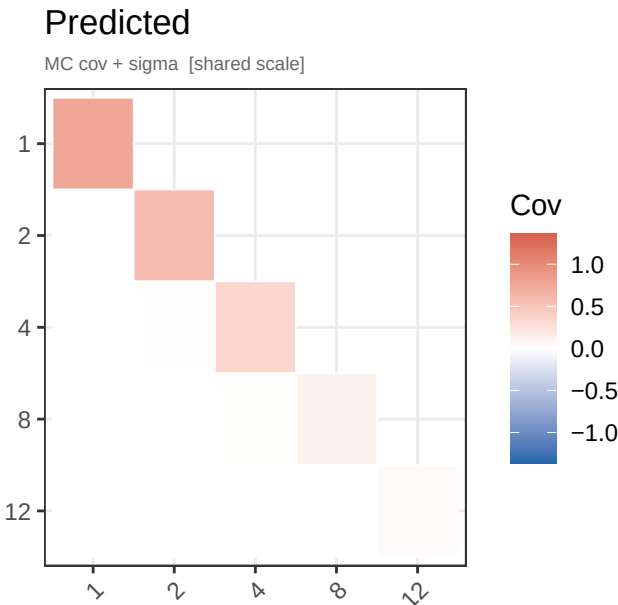
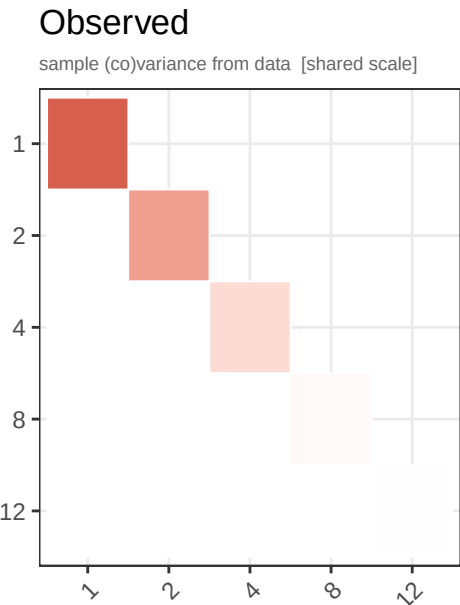


Parameter trace per restart

Natural scale; struct = back-transformed, sigma = SD, V(eta) = variance, off-diagonal = Cholesky $L[i,j]$



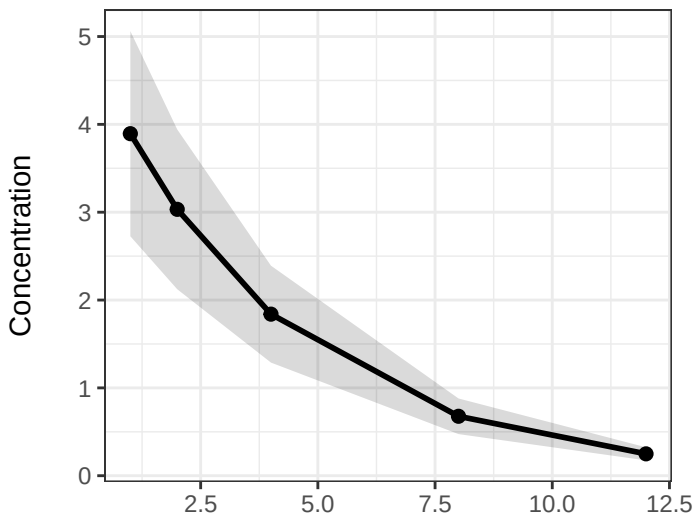
Study 's' -- Covariance diagnostics



Study 's' -- Mean diagnostics

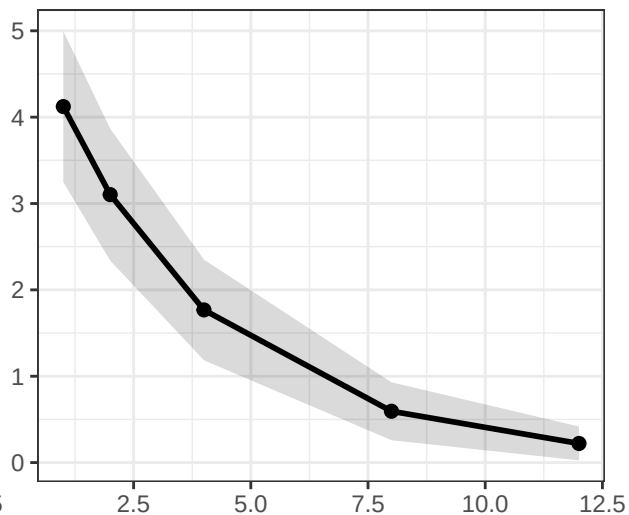
Observed

ribbon ± 1 SD [sqrtdiag(V_obs)] | shared y scale



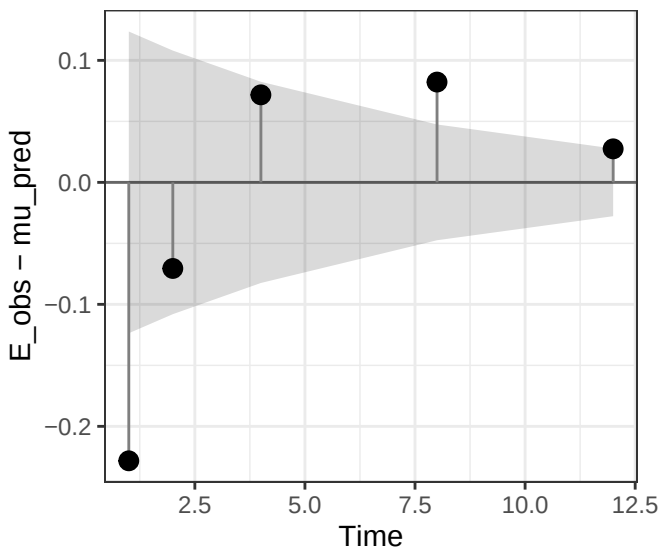
Predicted

ribbon ± 1 SD [sqrtdiag(V_pred); BSV + sigma] | shared y scale



Residual

band ± 2 SE(mean) [SE = sqrt(V_pred[t,t]/n)]



Standardised residual

dashed ± 1.96 | $z = \text{resid}/\text{SE}$ | * $p < .05$ ** $p < .01$ *** $p < .001$

